

Calibration yaw shifts narrow objects out of their image boxes

K4 · Track 4 · Day 6 | 3D from Point Clouds

CLAIM

On KITTI frame 000011, a 1° yaw error lowers the all-class hit ratio from 99.45% to 77.44% (-22.01 percentage points).

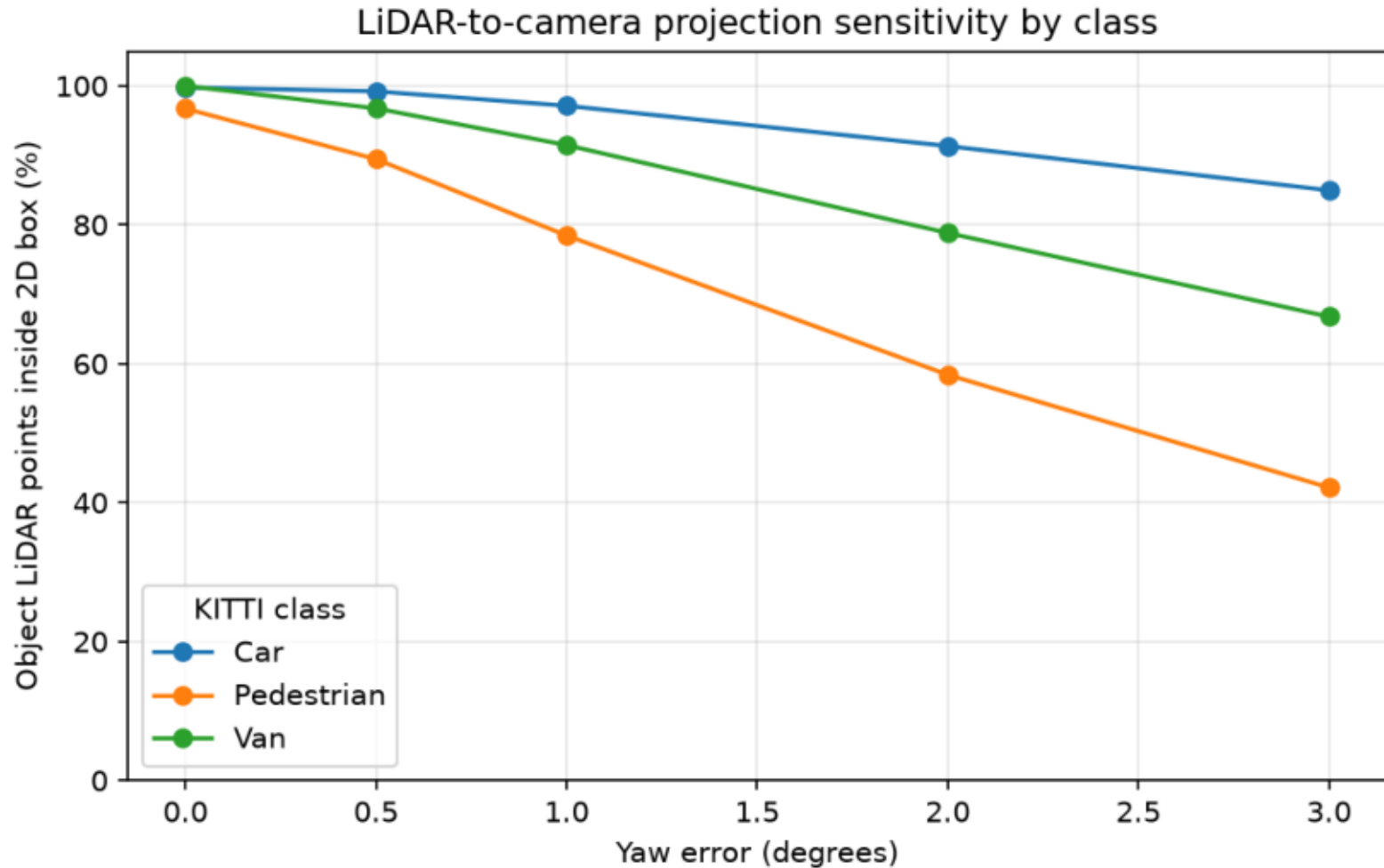
DATA

KITTI mini · frames 000008 (cars), 000011 (pedestrians), 000049 (occlusion)

Controlled variable: yaw error 0° , 0.5° , 1° , 2° , 3° .

Evidence: pedestrians lose alignment faster than cars

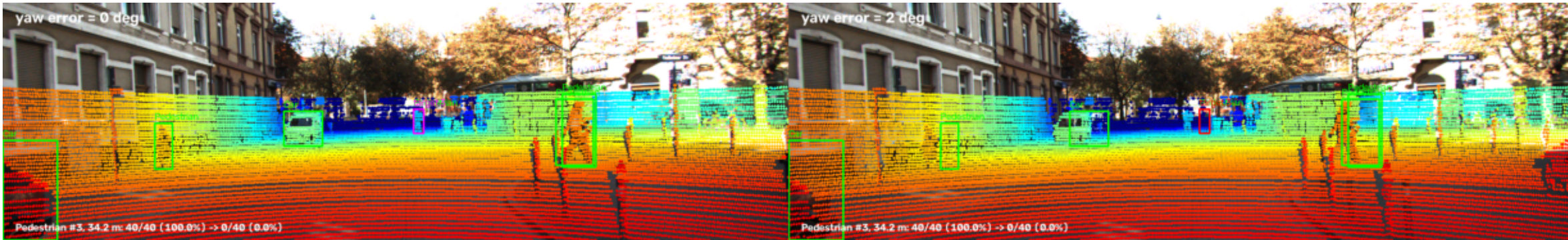
K4 · Track 4 · Day 6 | 3D from Point Clouds



At 1°: frame 000011 overall = 77.44%; frame 000008 Car = 98.62%.

Failure case: a 34.2 m pedestrian loses every in-box point

K4 · Track 4 · Day 6 | 3D from Point Clouds



Frame 000011 · yaw 2° · target #3: 40/40 points (0°) → 0/40 points (2°).

Debug layer: Geometry — LiDAR-camera extrinsic is misaligned.

Deployment recommendation

K4 · Track 4 · Day 6 | 3D from Point Clouds

Use case

Urban delivery vehicle; validate alignment at startup and after impact or strong vibration.

Log

Camera/LiDAR hit ratio by class and distance, estimated yaw, and sensor-mount temperature.

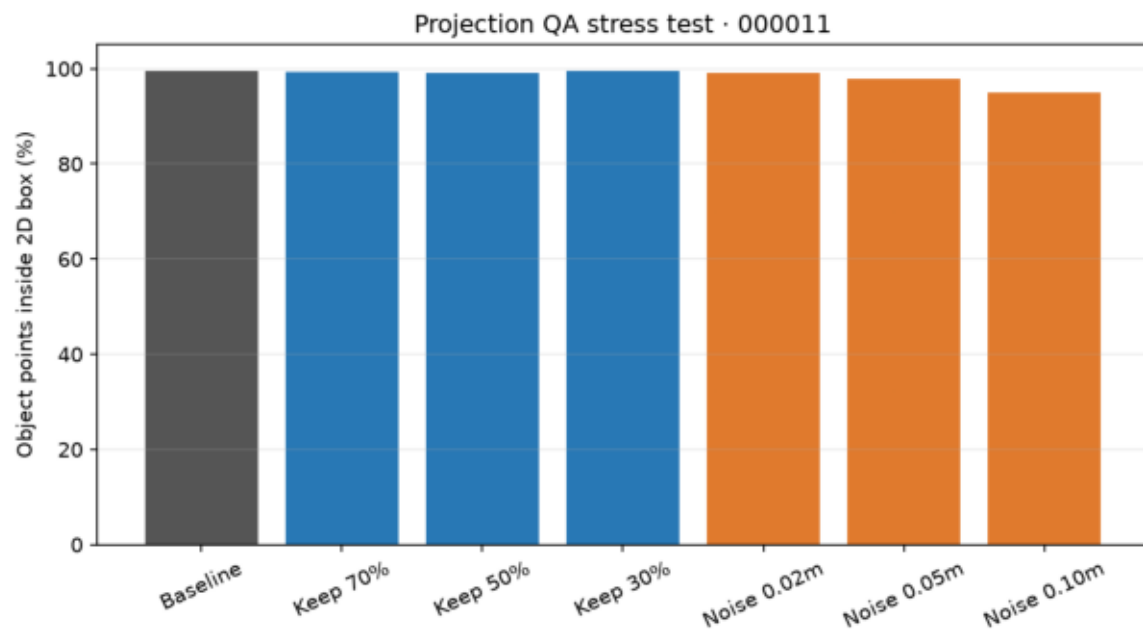
Alert

Below 90% for three consecutive checks. Run while stopped to limit runtime cost; require visible objects.

Study limit: three selected KITTI frames; the threshold needs validation on the target vehicle and camera detector.

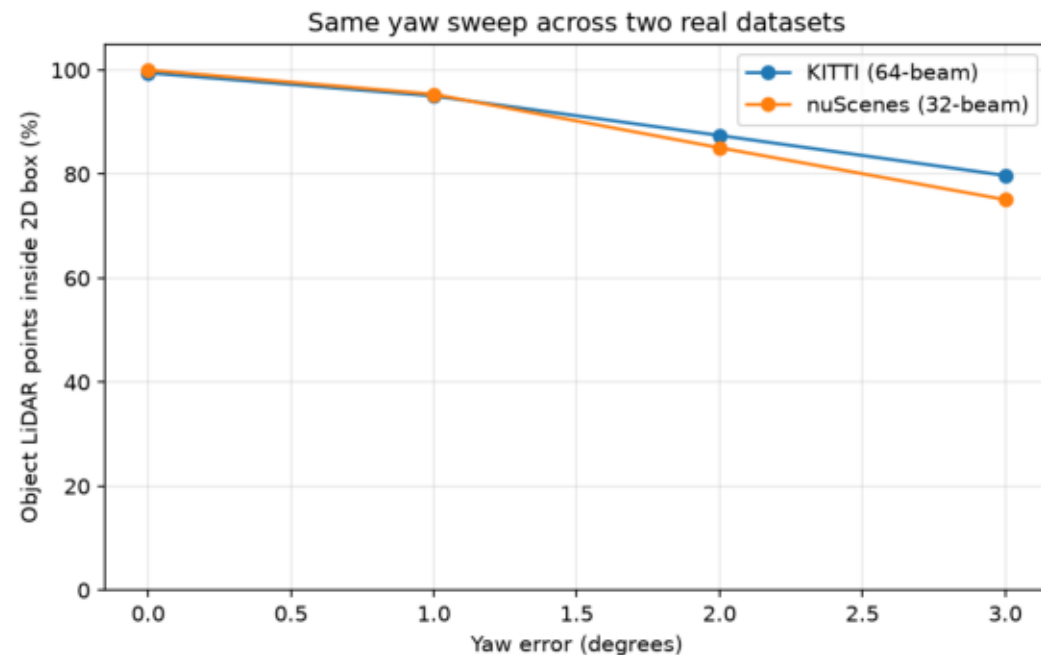
Bonus evidence: stress, latency, and dataset transfer

K4 · Track 4 · Day 6 | 3D from Point Clouds



B2 · dropout + Gaussian noise

Baseline 99.45%; 0.10 m noise 95.08%.



B5 · same yaw sweep, two datasets

At 3°: KITTI 79.73%; nuScenes 75.08%.

B3 · 20 timed runs after warm-up: p50 15.54 ms, p95 17.14 ms · Core i9-13900H · 15.6 GB RAM